AP Computer Science A

Course Outline • Mathlify Summer Program

Course Overview

This course prepares students for success in AP Computer Science A by building strong Java fundamentals, developing problem-solving skills, and practicing AP-style multiple choice and free-response questions. The goal is to ensure students enter the school year confident, prepared, and set up to earn an A.

Course Units

Unit 1 — Introduction to Programming with Java

- Java syntax, structure, and program flow
- Variables, data types, and expressions
- Basic input/output and debugging techniques

Unit 2 — Using Objects

- Creating and using objects
- Calling methods and understanding parameters
- Using the Java API and documentation

Unit 3 — Boolean Expressions & Conditionals

- Relational and logical operators
- If / else statements and nested conditionals
- Writing clear decision-making logic

Unit 4 — Iteration

- While loops and for loops
- Loop tracing and common patterns
- Applying iteration to problem solving

Unit 5 — Classes

- Defining classes and instance variables
- Constructors, methods, and encapsulation
- Object-oriented design principles

Unit 6 — Arrays

- Creating and traversing arrays
- Array-based problem solving
- Common AP-style array patterns

Unit 7 — ArrayLists

- Using ArrayLists and their methods
- Comparing arrays vs ArrayLists

• AP-style questions involving collections

Unit 8 — 2D Arrays

• Creating and accessing 2D arrays

• Nested loops for traversal

• Row-major and column-major order